MILITARY SPECIFICATION

GRENADE, HE, G.P., M42
LOADING AND ASSEMBLING

This Amendment forms a part of Military Specification MIL-G-50975C (AR), dated 4 January 1982, and is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1, under US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND, add the following:

"PRODUCT AND PACKAGING DRAWINGS

9349337 - Packing and Marking for Box Outer for Grenade: M42, M46 and XM77."

PAGE 3

3.2.1, add the following new paragraph:

"3.2.1 Lead cup assembly. The lead cup assembly shall be free of all splits or cracks."

PAGE 4

4.3.1, add the following first article sample items:

"Packable slider lock 9329680 30"

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4.3.2, Table I: Under the heading of Body Loading Assy, delete "3.2.1" and substitute "3.2" and delete "4.5.5.1" and substitute "4.5.5" for unseating force of cone swage. Delete "4.5.5.2 or 4.5.5.3" and substitute "4.5.5" for test for presence of caked explosive entrapped in groove.

FSC 1330

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Add the following inspection:

"Packable slider lock (Dwg. 9329680, Examination for defects per 4.4.2.9 3.0 3.2"

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4.4.2.4, Critical Defect: Add the following:

"2. Lead cup assembly split or cracked 100% 3.2.1 Visual".

Majors 101 and 102: Delete "4.5.5.1" and "4.5.5.2/4.5.5.3" and substitute "4.5.5".

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4.4.2.6, add Critical Defects 3 and 4 as follows:

"3. Slider lock cracked, incomplete, missing, or improperly positioned 100% 3.2 Visual
4. Safety clip of fuze missing, damaged, or double end not protruding beyond end of fuze tunnel NOTE (c) 100% 3.2 Visual".

Delete Major Defect 103, and add Major Defects 114, 115, 116 and 117 as follows:

"114. Height from the shoulder of grenade body to bottom of wound ribbon (two places) .40% 3.2 Gage
115. Presence of moly disulfide on phosphate coated slider lock .40% 3.2 Visual
116. Pull test, max. .40% 3.2 4.5.9
117. Pull test, min. NOTE (d) 3.2 4.5.9"

Add NOTE c and d as follows:

"NOTE c: Applies only if slider lock is not assembled immediately after tape winding operation.

NOTE d: The samples from Major 116, accept on 0, reject on 1."

4.4.2.7, 4.4.2.8, and 4.4.2.9: Add the following new tables:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Packing and Marking for Box, Fiberboard for Grenade, GP: M42, M46 and XM77		9349336
			NEXT HIGHER ASSEMBLY 9349337

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Partitions missing		0.40%	3.2	Visual
102	Tubes missing		0.40%	3.2	Visual
103	Barrier bag torn, punctured or heat seal incomplete		0.40%	3.2	Visual
104	Box damaged, exposing contents		0.40%	3.2	Visual
105	Corners not blunted		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Box marking missing, incomplete or illegible		0.65%	3.2	Visual
203	Barrier bag marking missing, incomplete or illegible		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349337
4.4.2.8	Packing and Marking for Box, Outer for Grenade: M42, M46 and XM77				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents loose		0.40%	3.2	Manual
102	Box damaged, exposing contents		0.40%	3.2	Visual
103	Strapping missing (wood box only)		0.40%	3.2	Visual
104	Seal missing (wire bound box only)		0.40%	3.2	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50975C (AR)
AMENDMENT 3

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Packable Slider Lock		9329680
			NEXT HIGHER ASSEMBLY 9215340

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Locking tab cracked, incomplete or missing		100%	3.2	Visual
Major					
101	Profile of minimum interior dimensions undersize		0.40%	3.2	Gage
102	Location of locking tab		0.40%	3.2	Gage
103	Length of locking tab		0.40%	3.2	Gage
104	Location of safety pin hole		0.40%	3.2	Gage
105	Width of safety pin hole		0.40%	3.2	Gage
106	Length of safety pin hole		0.40%	3.2	Gage
107	Pull test, max.		0.40%	3.2	4.5.9
108	Pull test, min.	(a)		3.2	4.5.9
109	Hardness test	(b or c)		3.2	4.5.10
Minor					
201	Width		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Protective coating incomplete or missing		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
(a) The same sample from Major 107, accept on 0 - reject on 1.
(b) 25 samples per 4 hours production, accept on 0 - reject on 1, by the producer.
(c) When alternate material (Martensite steel) is used, apply special inspection level S-4, AQL 0.15%. Manufacturer's certification of hardness for preheat treated material is required.

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4.5.4.5: Delete the last sentence "If the fuze loosens as evidence by a gap..." and replace with the following:

"The sample shall be removed from the fixture and inspected for a gap around the studs between the fuze housing and the cover, and between the fuze cover and the grenade cap. A separating force shall be applied manually to the fuze grenade assembly while gaging. If the sample exceeds the gap permitted on the applicable drawing, the sample assembly shall be classed defective".

4.5.5 and 4.5.5.1: Delete in their entirety and substitute the following:

"4.5.5 Unseating force test of cone swage and test for presence of entrapped powder of body loading assembly.

4.5.5.1 Preparation of sample items and test procedure for unseating force. The sample assemblies shall be randomly selected from normal production.

The quantity of body loading assemblies shall be cut in two in a plane perpendicular to the longitudinal axis of the body. The cut shall be located at a minimum distance of .7000 inches from the open end. The cut shall be made cleanly through the body wall and explosive material.

The portion of the cut sample containing the swage of the cone shall have any remaining explosive material cleaned away and then shall be mounted in an approved fixture. The load as specified on the applicable drawings shall be applied to the cone. If the cone is dislodged from its swaged position in the body loading assembly by the specified load, that body loading assembly shall be classed defective."

4.5.5.2 thru 4.5.5.5: Delete in their entirety and substitute the following:

"4.5.5.2 Test for presence of entrapped powder in body loading assembly. The test may be accomplished by one of the following procedures.

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a. Alternate 1. The sample assemblies shall be randomly selected from normal production. The quantity of body loading assemblies shall be cut in a plane perpendicular to the axis of the body at a minimum distance of .700 inches from the open end. The cut shall be made cleanly through the body wall, cone and filler material.

The ring containing the base end of the cone shall then be cut into two semi-circular halves so that the cone can be lifted out. An alternative to the above method is to cut the cone into two semi-circular halves leaving the base end of the body in ring form. The cone halves shall then be removed but without loss of any entrapped powder.

The two halves of the cone shall be lifted out and the body shall be visually examined for evidence of caked powder entrapped in the crimping groove (see 4.5.5.4).

b. Alternate 2. The sample assemblies shall be randomly selected from normal production. The quantity of body loading assemblies shall be cut on the longitudinal axis in two halves. The cut shall be made cleanly through the body wall, the explosive material, and cone. The sectioned cone in each half shall be lifted out and the body shall be visually examined for evidence of caked powder entrapped in the crimping groove (see 4.5.5.4).

4.5.5.3 Combined testing of force test and test for entrapped powder (alternative method). The force test and test for entrapped powder may be combined, provided the contractor uses powder nests or equivalent loading aids to reduce the occurrence of entrapped powder during grenade body loading. To use this alternative the contractor must, prior to use, request approval from the procuring agency and furnish documentation of the method planned to reduce the possibility of entrapped powder. For this combined test the same samples are used for both the force test and entrapped powder test. After having passed the 2000 lb. force requirement, the load shall be increased until the cone is dislodged from the grenade body. The grenade body shall be examined for evidence of powder that had been entrapped in the crimping groove (see 4.5.5.4).

4.5.5.4 Reject criteria of the test for presence of entrapped powder in body loading assembly. The body loading assembly shall be cause for the rejection, if the test reveals any one of the following:

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a. The explosive material fills the crimping groove between upper and lower swaging ridge in excess of 30 degrees of the circumference and it is either continuous or additive.

b. The presence of explosive material in the crimping groove is in excess of 90 degrees of the circumference and it is either continuous or additive.

4.5.5.5 Disposition. If any body loading assembly fails to comply with any of these tests the production quantity represented by the test sample shall be rejected."

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4.5.9 and 4.5.10: Add the following new paragraphs:

"4.5.9 Pull test of slider lock (non-destructive). Each sample slider lock shall be assembled to a fixture simulating a maximum dimension fuze housing and pull tested until the slider lock releases. The load shall be applied to the top of the lock in a direction away from the simulated housing. Any slider lock failing to meet the force requirement on the applicable drawing shall be classed defective.

4.5.10 Hardness test of the slider lock (non-destructive). This test shall be performed in accordance with ASTM-E-18 using the standard penetrator for the specified hardness scale. Readings shall be taken on the top surface of the slider lock. Any slider lock failing to meet the applicable hardness requirement shall be classed defective."

5, delete the parenthetical note in its entirety and substitute the following:

"5. PACKAGING (This section is not applicable to intraplant usage)

5.1 Packaging. Level A (not intended for long term storage) and Level B: Packaging shall be in accordance with drawing 9349336.

5.2 Packing. Level A (not intended for long term storage) and Level B: Packing shall be in accordance with drawing 9349337.

5.3 Marking. Marking of fiberboard box and bag shall be in accordance with drawing 9349336. Outer box shall be marked in accordance with drawing 9349337."

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6.4, add the paragraph column, "4.5.9 and 4.5.10" for Quality Inspection Test Demonstration and Evaluation Report.

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